Physical activity detection in elderly diabetes patients using raw smartwatch data

Bachelor End Project Report

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Abstract

Physical activity (PA) is crucial for maintaining a healthy lifestyle and managing various health conditions (e.g., diabetes mellitus). However, accurately detecting PA in free-living conditions from accelerometer data remains a challenge. Currently, the cut-points approach is widely used in PA research, but it requires resource-expensive calibration studies. This study presents a more patient-specific, objective unsupervised machine-learning method for automatic PA detection using wrist-worn accelerometers. Hidden Markov Models (HMM) were implemented to detect PA in data collected from 11 elderly diabetes patients. The performance of the HMMs was compared with self-reported activities and a traditional cut-points approach. The HMMs (HMM[acc] and HMM[acc+angles]) showed promising results with average recall rates of 0.71 and 0.61 respectively, outperforming the standard cut-points approach with a recall rate of 0.22. However, both HMMs displayed severe overestimations and inconsistencies in the predicted minutes spent in PA in comparison to the participant-reported activities. Still, HMMs could provide a valuable step towards objective physical activity detection, without the need for resource-expensive calibration studies, enabling a better understanding of the impact of physical activity on health and disease.

1 Introduction

Current guidelines recommend that people engage in at least 150 minutes of moderate activity per week or 75 minutes of vigorous activity. These guidelines are supported by years of research showing that active individuals have reduced risks of cardiovascular disease, cancer, and mortality compared to inactive individuals [1, 2]. Engaging in physical activity (PA) is also recommended for people with diabetes mellitus. Diabetes is a chronic metabolic disorder that occurs when the body is unable to produce insulin, or the body produces insufficient insulin, to maintain glucose homeostasis. Both lead to elevated blood sugar levels. According to the International Diabetes Federation (IDF), approximately 537 million adults had diabetes in 2021, and this number is expected to exceed 700 million by 2045 [3].

Exercise can help people with diabetes attain the target blood sugar levels, lipid profile and reach their desired body composition[4]. However, it is important to note that there may be inter-subject variability in the short-term effects of exercise on blood glucose levels. Research into these effects as well as capturing and forecasting the effects of PA on patient-specific blood glucose levels require objective and accurate PA detection. Previous research into the effects of PA on blood glucose levels used logbooks of time spent in PA reported by the participants, but self-reported measures are subjective and notably overestimate time spent in PA[5].

Smartwatches with accelerometers offer a promising solution for collecting continuous and patient-specific data. However, the development of algorithms to accurately detect PA from smartwatch data remains a challenge. Recent studies have explored various machine-learning techniques to classify physical activities based on smartwatch sensor data. For instance, Montoye et al. [6] performed cross-validation and out-of-sample testing to predict PA intensity using wrist-worn accelerometers. Ballı et al. [7] proposed a hybrid approach combining principal component analysis and the random forest algorithm for human activity recognition from smartwatch sensor data. However, most studies were conducted in lab settings, not in free-living conditions. In lab settings, the performed PA is usually not representative of the PA performed without a protocol and supervision[6].

A recent study conducted by the TU/e and the Máxima Medical Centre study asked elderly diabetes patients to, among others, wear a Samsung Galaxy Watch 2 Active in free-living conditions for 14 days[8]. The participants were also asked to report their activities during that time. The data collected from this study is more representative of PA performed in free-living conditions than data collected in lab settings from other studies. The most commonly used method for detecting PA is the cut-points approach, which allows the calculation of time spent in PA using the acceleration by setting thresholds[9]. However, the complex relationships of acceleration with activity types, study populations, and study designs, make the cut-points approach easily overfit the experimental conditions from which the thresholds are derived. Since the corresponding physical intensity of a measured acceleration is specific for each population and accelerometer. Conducting resource-expensive calibration studies for all populations and activity types is not possible, thus calling for a new approach. For the new approach, training a model on the logbook data is not possible since this would generate a model that predicts the logbook reports rather than the actual time spent in PA. For that reason, an unsupervised learning method, which learns patterns and structures from unlabeled data, is needed.

This research aims to implement a patient-specific objective unsupervised machine learning method for the automatic detection of PA using wrist-worn three-axial accelerometers. The model must be able to detect bouts of PA instead of labeling singular data points. To achieve this the primary aim is to implement a Hidden Markov Model (HMM). HMMs are stochastical models used to describe systems that are assumed to be Markov processes with hidden states[10]. The performance of the HMMs is compared with the self-reported activities as well as a more traditional cut-points approach, using different performance measures. It is expected the HMM will show more consistency across participants in detecting physical activity, as the HMM tailors the detection method to each individual. An additional goal was to evaluate the performance of the HMM using accelerometer orientation metrics, along with acceleration, as input.

2 Method

2.1 Data

Data was used from the DiaGame study, where 11 elderly patients with diabetes (10 with type 2 diabetes, 1 with type 1 diabetes) were instructed to wear a commercially available Samsung Galaxy Watch 2 Active on their wrist during the day[11]. This watch measured the acceleration for the three axes relative to its body frame, the heart rate, and pedometer data (this includes steps taken, moved distance, and calories burned). These measurements were conducted for 14 days in free-living conditions and no instructions were provided that would alter their day-to-day lives or habits since it is an observational study.

The accelerometer measures the acceleration on three axes relative to the watch's body. The values measured are in $\mathrm{m/s^2}$ and the device at rest should measure the acceleration of gravity on one of its axes (1g / $9.81\mathrm{m/s^2}$). The watch measures with an inconsistent sampling rate since the smartwatch can go into hibernation mode to save energy. The specific details of the variations in sampling rate are not provided by the manufacturer (Samsung). During these periods pedometer data may be accumulated, accumulation is not possible for acceleration data and heart rate data.

To preprocess the data and to focus on the relevant measurements certain steps had to be taken. First, samples outside of the 14-day period had to be removed, as well as all the negative heart rate measurements. These include samples where the watch could not measure a heart rate, or the watch was not even worn. The participants were also asked to report their activities, including the activity details, start time, and estimated duration. Investigating those logbooks revealed that only 5 out of 11 participants reported any activities, so any comparison to reported activities only contains their data.

2.2 Feature extraction

For the relevant timestamps, the following features were defined. First, the vector magnitude of the acceleration is calculated as Euclidian Norm Minus One (ENMO)[9]. The measured acceleration is set to gravitational units by dividing it by 9.81(1g). Then by calculating the Euclidean norm and subtracting one gravitational unit, the ENMO values are computed. This provides a measure of the body's acceleration.

$$\text{ENMO} = \sqrt{\left(\frac{\text{acc}_x}{9.81}\right)^2 + \left(\frac{\text{acc}_y}{9.81}\right)^2 + \left(\frac{\text{acc}_z}{9.81}\right)^2} - 1 \tag{1}$$

Where acc_x, acc_y and acc_z refer to the acceleration pointing in their respective direction. Negative values that are computed, indicate that the watch may not have measured the gravitational component accurately. To account for this all negative values are set to zero. By setting it to zero, it still counts as a measurement and has an influence on the ENMO model rather than discarding the whole measurement and biasing the result[12]. Then the orientation angles of the acceleration vector are calculated, since patterns or ranges of angles may correspond to activities[9]. Using the following formulas, where the units of the resulting angles are degrees:

$$\text{angle}_x = \tan^{-1}\left(\frac{\text{acc}_x}{\sqrt{\text{acc}_y^2 + \text{acc}_z^2}}\right) \cdot \frac{180}{\pi} \tag{2}$$

$$\text{angle}_y = \tan^{-1}\left(\frac{\text{acc}_y}{\sqrt{\text{acc}_z^2 + \text{acc}_x^2}}\right) \cdot \frac{180}{\pi} \tag{3}$$

$$\text{angle}_z = \tan^{-1}\left(\frac{\text{acc}_z}{\sqrt{\text{acc}_x^2 + \text{acc}_y^2}}\right) \cdot \frac{180}{\pi} \tag{4}$$

2.3 Gaussian Smoothing

To remove extreme values and noise from the features, a Gaussian filter is applied. This could help with enhancing signal quality, improving accuracy, and making the data consistent and comparable. The Gaussian filter is suitable since it is important to preserve the underlying signal, and it is an easy-to-interpret, well-studied filter. For every timestamp the Gaussian kernel values are calculated based on the time difference, using the following formula:

$$K(t, t_i) = \exp\left(-\frac{(t - t_i)^2}{2\sigma^2}\right) \tag{5}$$

Where t is the target timestamp, t_i represents a neighboring timestamp, and σ is the standard deviation. This standard deviation determines the width of the kernel and how much the other timestamps influence the kernel value. To smooth out active instances in inactive periods and account for measurement errors, σ was set to $\frac{5}{6}$. The filter was truncated at t - 2.5 minutes and t + 2.5 minutes to speed up computation times as these values are not expected to have considerable influence on the value of interest. The choice of kernel size is a trade-off between smoothing and preserving the details of the data. A larger kernel size can lead to too much smoothing and valuable information might be lost. This kernel has the total size of the minimal time for an activity in the ENMO method, to lower the chance of losing an activity. Making the kernel smaller showed that some extreme values were still present after applying the filter.

After calculating the appropriate kernel values, it is necessary to normalize them to ensure that their sum equals one. Let's denote the kernel values for an instance as $K(t^*)$, and the corresponding feature values in that window as $F(t^*)$. Each kernel value is divided by the sum of all kernel values:

$$\text{Normalized kernel values: } N(t^*) = \frac{K(t^*)}{\sum_i K(i)} \tag{6}$$

The normalized kernel values are then multiplied element-wise with the feature values:

$$\text{Weighted feature values: } W(t^*) = N(t^*) \cdot F(t^*) \tag{7}$$

The resulting weighted feature values are summed to obtain the new value for the feature at instance t:

$$\text{New feature value: } V(t) = \sum_i W(t^*) \tag{8}$$

When applying the Gaussian filter to the orientation axes independently the new signal did not have similar characteristics as can be seen in appendix 9, even when trying with smaller kernel sizes. So the filter is applied to the calculated features. By applying the filter afterwards the noise suppression is performed on the relevant transformed data.

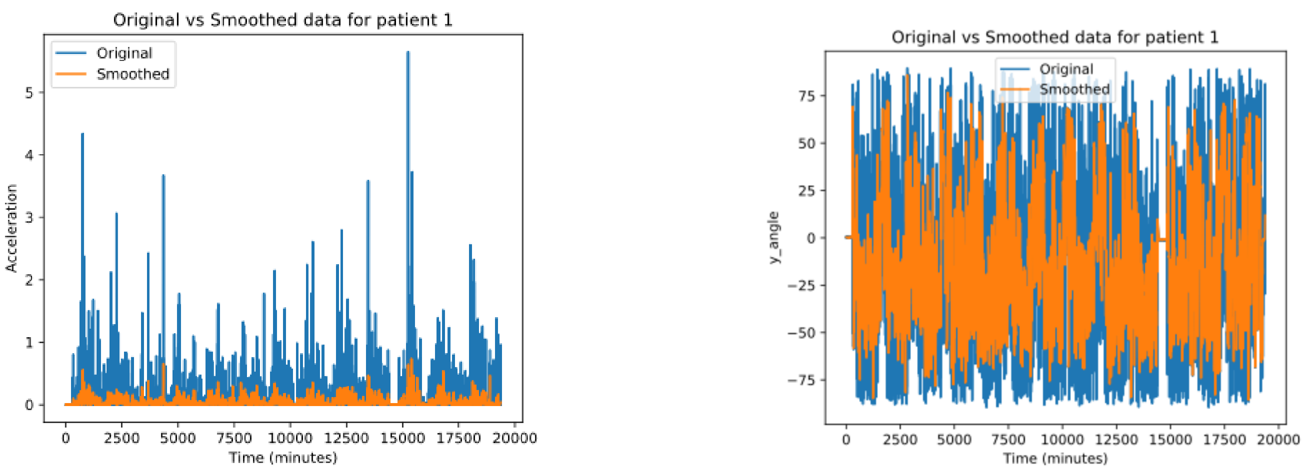


Figure 1: Original ENMO values and angle$_y$ plotted against their smoothed values for participant 1

2.4 ENMO model

Based on the acceleration and angles the cut-points approach was created to predict periods spent in PA. The model is built to detect prolonged bouts of activity (e.g., walking, cycling) and not just short bursts of activity such as sprints. Short bursts of activity are highly unlikely for the population in which the data was collected since most participants likely did not have the physical capabilities to engage in such vigorous activities. By targeting longer active periods, the model also aligns with recommendations and evaluation of compliance with the recommended health guidelines.

First, the model searches for a timestamp that is considered to be active which means the acceleration at that moment is above 0.04g[9]. This is the reported acceleration value for light PA from a study on children and adults[13]. After a starting point for a possible activity has been found it checks if $angle_z$ changes at a speed larger than 1 degree per second at least once in the next 5 minutes[12]. If this does not happen, the point is not considered to be the start of an activity and the model moves on to the next active point.

If in a 5-minute period, at least 90 percent of points are considered above the acceleration threshold and the test of $angle_z$ was passed then this period is considered active. The remaining 10 percent accounts for natural variations in acceleration values inside one activity. These short interruptions of inactivity are not allowed to last longer than a minute, even if 90 percent of timestamps are still above the threshold. If the interruption is shorter than one minute it is counted as time in activity and counted towards the total estimated active time in that day. If not, the model ends the activity and starts looking for a new activity. When the 5-minute period is considered to be active, the activity is extended with the following timestamps, until less than 90 percent of the points are above the threshold, or for 1 minute no new point was above the threshold[9].

2.5 Hidden Markov Model

A Hidden Markov Model is a stochastic model based on the idea that an observed time series has been generated by an underlying, unobservable process that has a random probability distribution, whose random states X_t are not directly observable[14]. This sequence of states satisfies the Markov property and forms a Markov chain. A Markov chain consists of a set of states and a set of transitions between those states. At any given time, the system is in one of the states. It starts with an initial probability matrix (π_0), which specifies the probabilities of the process to start in each of the existing states. The probability of transitioning to a future state depends solely on the present state and is unaffected by the history of the system before the current state.

$$P(X_{t+1}|X_t, X_{t-1}, \ldots, X_0) = P(X_{t+1}|X_t) \tag{9}$$

After the initial state, the probability of transitioning from state X_i to any other state X is contained in row i of transition matrix A. For a regular Markov model, the goal is to get the most likely time series of length T containing states $X_1, \ldots, X_T$, using the initial and transition probabilities. In a HMM the states can only be observed indirectly, so X_t is never given. However, X_t can be inferred using a parallel time series represented by the input of the smartwatch data, where each observation O_t of the input has a probability of being emitted from a state $i \in X$. The probabilities of an observation being generated from a state are contained in emission matrix B. An important assumption for the HMM is that the probability of an observation to be made in time t^* is only dependent on the state X_t and nothing before that:

$$P(O_t = o_t \mid X_1, \ldots, X_t, O_1, \ldots, O_{t-1}) = P(O_t = o_t \mid X_t = X_t) \tag{10}$$

In this case, the observed features are real-valued measurements, where for each state i the probability of an emission value b_i is derived from a Gaussian distribution with its state mean μ_i vector and covariance matrix Σ_i. This assumes that the data will be Gaussian distributed, which for continuous variables is most commonly assumed [15]. When working with several features for every observation, the distribution will be a multivariate Gaussian, where any linear combination of the random variables also follows a Gaussian distribution[16]. The covariance matrix of this distribution is diagonal, meaning that it only includes variances along the diagonal and assumes zero correlation between different features, this simplifies the estimation process. So given the emitted observations, the model will evaluate the probability that the Markov chain is in a state using the corresponding Gaussian probability distributions. Combining the emission probabilities with the initial probabilities and the transition probabilities the HMM will infer the most likely time series of length T containing states $X_1, \ldots, X_T$.

$$\begin{tikzcd}
X_0 \arrow[r, "A"] \arrow[d, "B"'] & X_1 \arrow[r, "A"] \arrow[d, "B"'] & \ldots \arrow[r, "A"] & X_m \arrow[d, "B"'] \\
\mathcal{O}_0 & \mathcal{O}_1 & \cdots & \mathcal{O}_m
\end{tikzcd}$$

Figure 2: Hidden states and their emitted observations with probability matrix B in a HMM[17].

The first part of the HMM is given the observed data, estimating the model parameters. This can be solved by the iterative Expectation-Maximization (EM) also known as the Baum-Welch algorithm. Its purpose is to tune the parameters of the HMM, the state transition matrix A, the emission matrix B, and the initial state distribution π_0, such that the model is most likely to match the observed data. It starts by randomly initialising the transition matrix, emission matrix, and initial state distribution. The algorithm then iteratively performs two steps: the Estimation-step and the Maximization-step. In the E-step, the algorithm calculates the expected hidden states based on the current set of parameters and the observed data. In the M-step, it updates the parameters of the HMM-based on the previously computed expected hidden states. The aim is to maximize the likelihood of the observed data given the estimated hidden states, i.e. make the HMM more representative of the true underlying process that generated the data. These two steps are repeated until changes in the likelihood become insignificant for each iteration, resulting in an optimized HMM that fits the observed data.

Since the Baum-Welch algorithm is an iterative process, when initialised with random parameters it could converge at a local optimum. Local optima represent solutions that are relatively good within a limited region of the parameter space but may not capture the best overall fit for the data. To prevent the model from converging on a local maximum the algorithm could be initialised with prior information. The mean μ_i and the covariance matrix Σ_i of the Gaussian emission distributions could be initialised based on observations X related to the state s_i or prior information from literature [18]. In this case, due to a lack of training data and comparable literature, initialisation using a grid search is performed. This will test how well the model fits the data for random initialisation and for initialisation around literature values, combined with a general look at the data distribution. By testing the goodness-of-fit around those values more general optimal initial parameters can be found.

To assess the goodness-of-fit of the HMM and its parameters, three measures are most commonly used, the Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), and Log-likelihood (LL)[19][20]. The first two parameters take into account the goodness-of-fit of the model and the trade-off between model fit and model complexity, while the LL is just used as a measure of the model's goodness-of-fit to the data. Overall this will show how well the HMM parameters capture the observed data, this does not say anything about its ability to predict the correct labels. After the model is fitted and the parameters are estimated, the Viterbi algorithm gives the most likely sequence of hidden states[21]. This can be interpreted as the prediction of the model for every known timestamp.

For this research two feature combinations are defined, a model with the ENMO values as the feature (HMM[acc]) and a model with the ENMO values as well as angle$_x$, angle$_y$, and angle$_z$ (HMM[acc] and HMM[acc+angles]). These models will be tested and compared to assess if the addition of angles might improve the model's ability to find patterns, this will make it interesting to test if it finds relevant patterns for detecting activities.

2.6 Validation

To test the performance of the models different validation measures are used. Firstly, to take away the error participants make in guessing the exact timestamps they were active, the model's total estimated minutes per day are compared to the reported total minutes on that day. Plotting and computing this comparison shows a general idea of how the model performs. To check if the model predicted realistic activities, also plots for the duration of an activity are made. This will show the difference between the methods and could show flaws in the models if extreme values are encountered.

Another score to quantify the accuracy of the model from the total minutes predicted per day is the Relative Mean Squared Error. The RMSE score calculates the average difference between the predicted and logbook-reported minutes per day. A lower score represents more resemblance with the logbook, since the predicted values are closer to the reported values on average.

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} y_i^2}} \tag{11}$$

Another variable to compare the performance of the models to is the heart rate of the patient. Every time somebody does exercise their heart rate will rise above rest level. Thus, if the heart rate was below the measured heart rate at rest, this time was probably not spent in PA. However, stress, alcohol, and medications can also cause heart rate elevation, so not every time their heart rate increases they are engaging in PA. The heart rate is still a useful measure for testing the performance, but it cannot be seen as ground truth.

Confusion matrices can also give an insight into the behaviour of the model. The matrix consists of four quadrants: true positives (correctly predicted positive instances), true negatives (correctly predicted negative instances), false positives (incorrectly predicted positive instances), and false negatives (incorrectly predicted negative instances). Periods where activities were reported were most likely spent in activity. Thus, a good model should not have many false negatives. The recall rate shown in equation 12 will tell how many true positives get predicted out of all the positives in the dataset. Other possible measures might be influenced too much by possible errors in the reported activities, since reported bouts of inactivity data cannot be assumed to be negative instances.

$$Recall = \frac{TruePositive}{TruePositive + FalseNegative} \tag{12}$$

3 Results

3.1 Model Optimization

To determine the most likely number of states within the HMMs, the distribution of the ENMO values was assessed, taking into account the assumed Gaussian emission distributions. The distribution gives information about the underlying characteristics and structure of the acceleration. The number of visible peaks can imply something about the number of modes or clusters in the data. This information can be used to infer the right number of states to choose for the HMMs.

Since the HMM assumes a Gaussian distribution within the states for the features, it is important to make sure the data closely represents a Gaussian shape. Figure 3 shows that the ENMO values were heavily skewed towards zero and the values were non-negative. A logarithmic transformation can transform non-negative and skewed data so that it follows a normal distribution [22]. Setting all zero values to an extremely small number beforehand (10^{-6}) and then transforming the acceleration values to log space also showed this as can be seen in Figure 3. The zero values will not influence the shape of the overall curve but are also not discarded to prevent bias. Since most plots showed three peaks and some showed less, as can be seen in appendix 10, for the HMMs it was chosen to differentiate between three states and use the log-transformed ENMO values as input.

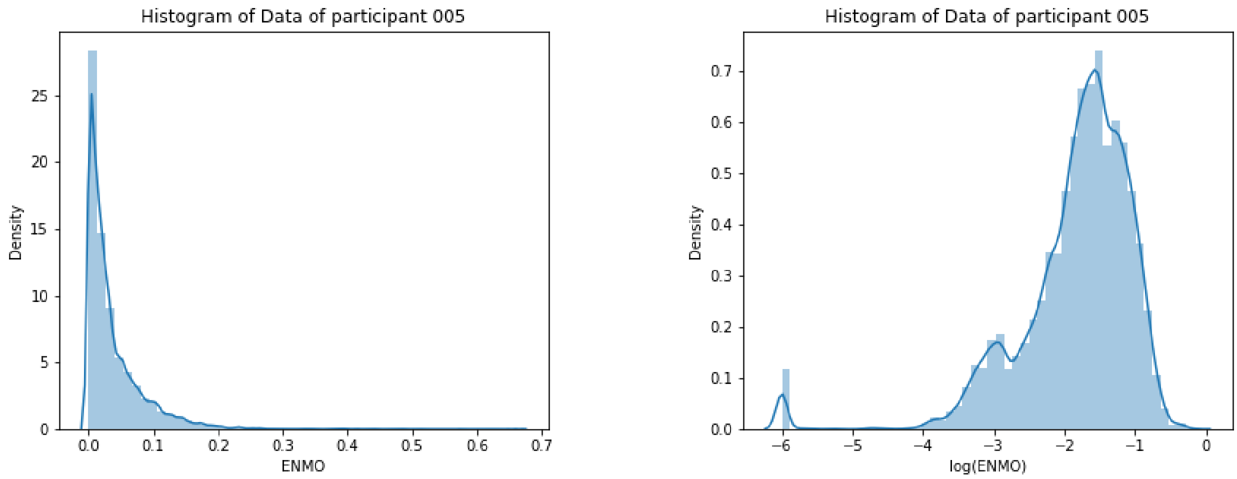


Figure 3: Original compared to the logarithmic distribution of the ENMO values for participant 5

After the data was log-transformed and the number of states are set to three, it is important to find the best initialisation method. Figure 4 shows the goodness-of-fit for 25 tries of complete random initialisation for HMM[acc], this was done for all participants. Visible in the plot is a large difference in the goodness-of-fit for every try. The goodness of fit can be three times higher for different tries, which shows that a random initialisation can give completely different models.

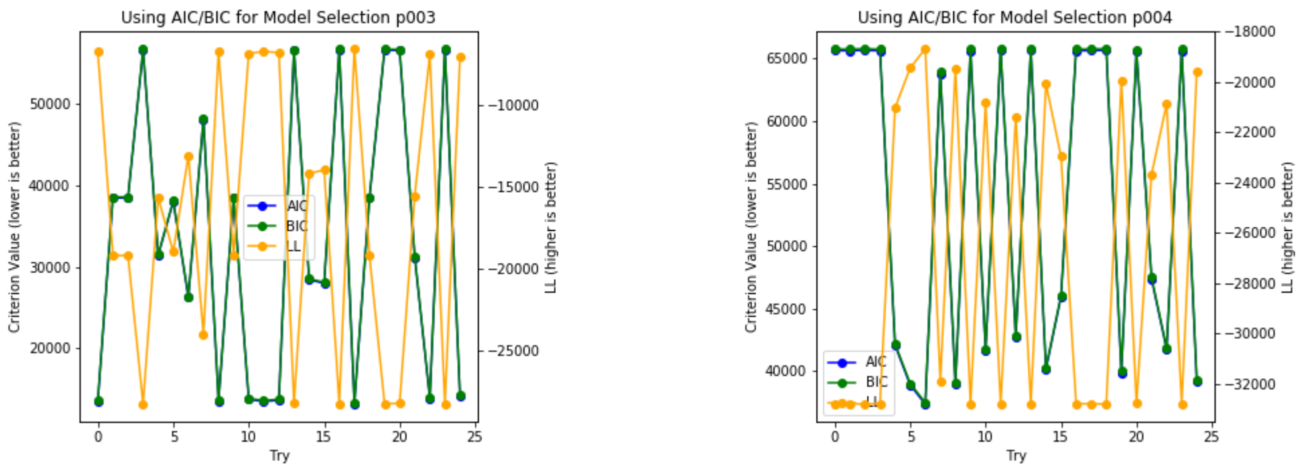


Figure 4: Goodness-of-fit with different random initialisations for HMM[acc]

It was observed that the goodness of fit is heavily dependent on the initialisation of the model, calling for a need for prior knowledge to initialise the model so it produces more consistent results. No prior knowledge about the initial state or the transition matrix is known. The initial means and initial co-variances of the Gaussian emission distribution do have this prior knowledge available, which could help the model converge in the correct optimum. From literature starting values could be deduced[9], and for the histogram plots it was observed that most peaks were visible in the plot around -6, -3 and -2. This implies that the mean of the log-transformed ENMO values for the different modes could be around those values. After trial and error testing by setting the means around those values and setting the covariances based on literature a more robust initialisation was found. This gave better and more consistent results for each participant, visible in Table 1. Testing the goodness-of-fit for these initial means and covariances showed a more consistent goodness-of-fit,

Table 1: The means and covariances of the Gaussian distributed emission probabilities with which the model was initialised.

State	Mean	Covariance
0	-3.07214526	1.07
1	-2.00274243	0.058
2	-1.21656154	0.061

but it is still not perfect, since in some instances the LL score drops to twice as low as its best score, seen in Figure 5. This difference can come from the fact that the initial probability matrix and the transition matrix are not explicitly initialised and still could converge to different optima.

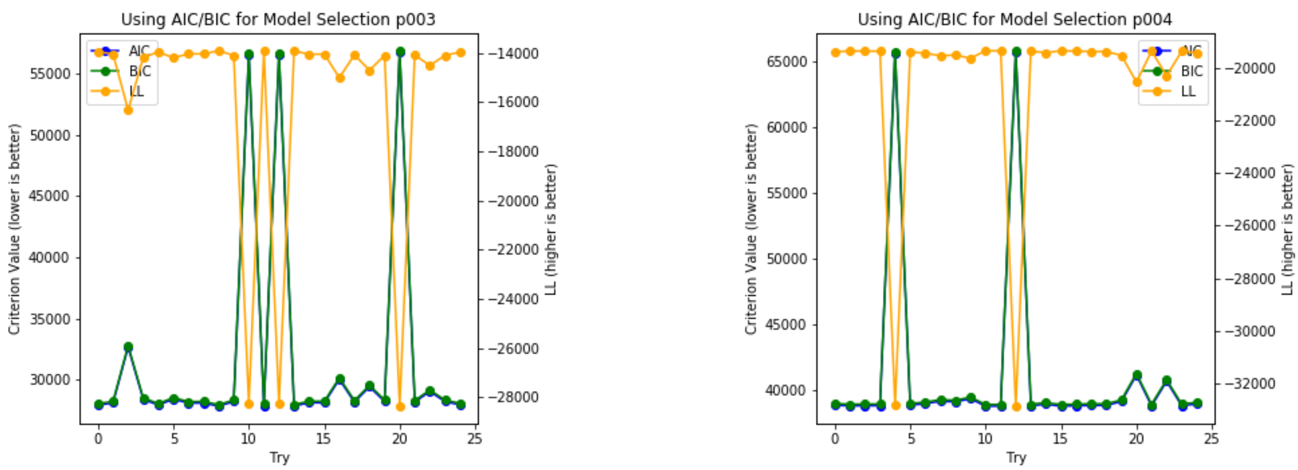


Figure 5: Goodness-of-fit with initialisation of prior from Table 1 for HMM[acc].

No general initialisation was found for initialising HMM[acc+angles], but this method also showed less overall dependence on the initialisation values. The goodness-of-fit showed a maximum variation of 5% for the random initialisation, visible in appendix 11, meaning that this was less dependent on the prior than HMM[acc]. To promote the clear distinction of the Gaussian distributions for the ENMO values, the same means were used to initialise HMM[acc+angles] as for HMM[acc], while the angle means and covariances of the Gaussian emission distribution were randomly initialised.

To make the step from predicted states by the HMMs towards actually detecting activities, the state with the highest mean acceleration was chosen as the active state and the two states with the lowest means are seen as inactive. The original values of the lowest two means generally are around 0.00032g (after log-transformation this was -3.5) and 0.001g (-2) which is far below the threshold of 0.04g (-1.39) from the ENMO method and the highest mean usually is around 0.05g (-1.3), which is just above the activity threshold. The final parameters the models converged to for different participants can be found in the appendix 5.

3.2 Model Comparison

The three methods (ENMO model, HMM[acc], HMM[acc+angles]) are compared to the logbooks kept by the participants. Figure 6 shows the spread of the duration of activities predicted by the corresponding model. The plot shows extreme values for participants 1 and 7, in the plot next to it these participants are removed to show more detail. Visible from the plots is that the durations of reported activities differ greatly per participant, but there is no clear visible relation in the way the models differ. The duration of logbook-reported activities is heavily dependent on the participant that reported them. The ENMO model has a minimum duration of 5 minutes for an activity which is also visible in the plot. For the HMMs it is apparent that the lower boundaries of the boxes are close to zero, and the median is below five minutes. Indicating that a lot of short activities were predicted. The upper whiskers of the box were relatively short for the ENMO model and there are a lot less visible outliers compared to the HMMs.

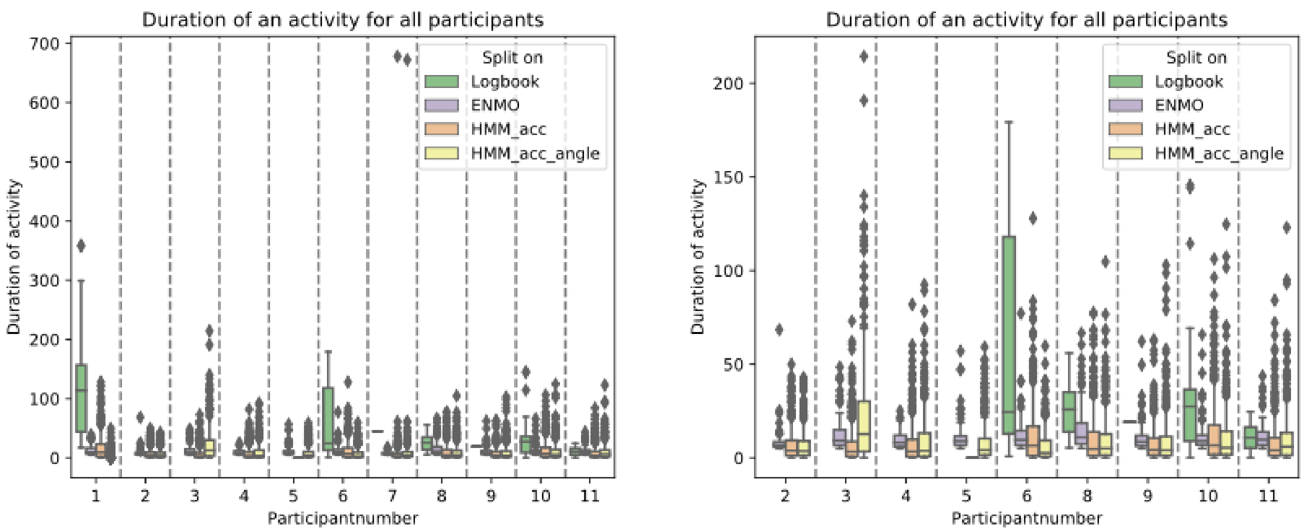


Figure 6: Boxplots of the duration of the predicted activities for each participant according to each method. The right plot has participants 1 and 7 removed.

A comparison day by day of how many minutes were estimated to be active shows the overall prediction behavior of the model, without the model getting penalized if the exact timestamps in the logbooks are reported incorrectly. It will also demonstrate if the predictions are realistic. In Figure 7 it is visible that the HMMs predicted a lot more minutes spent in PA compared to the ENMO model and participant-reported activities. It also shows that for participant 4 the HMM[acc+angles] consistently predicts more daily minutes in PA, while for participant 10 HMM[acc] consistently predicts daily minutes in PA. This is also visible in appendix 12, which shows all participants. Meaning that the predictive behaviour of both models differs per participant, but for that participant there is a relatively consistent daily difference.

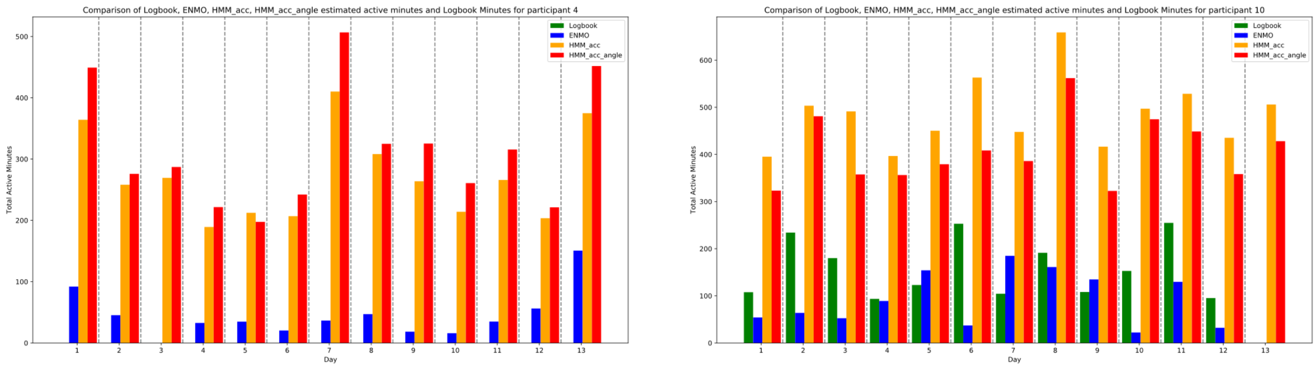


Figure 7: Predicted day-by-day minutes spent in PA comparison of different methods (see legend for associated colors) for participants 4 and 10.

For every day the ratio of predicted minutes and reported minutes was computed and for the corresponding participant, the mean and median of that ratio for the 14-day period are shown. Where the mean of the ratio's infinite values (days where no activities were reported by the participant) are not taken into account and the median infinite values are moved to the end, this way the infinite values are still accounted for as an overestimation. Table 2 also shows the Root Mean Squared Error which is a measure for the average magnitude of the differences between predicted and actual values, this measure also shows that the ENMO model shows the least difference with the daily minutes in activity that were reported. For the mean and median a score close to 1 implies the daily reported minutes for the model are very similar to the logbooks, the ENMO model is most consistently close to that. If the model fits the logbook data perfectly the RMSE will be zero, also for this measure the ENMO model shows better results. Thus, it is visible that the ENMO model is more in agreement with the daily reported minutes.

Table 2: Root Mean Square Error for each category as well as a measure for the ratio of predicted and reported minutes

	1			6			8			10			11		
	RMSE	Median	Mean	RMSE	Median	Mean	RMSE	Median	Mean	RMSE	Median	Mean	RMSE	Median	Mean
ENMO model	0.84	0.59	0.4	0.72	0.66	3.21	4.13	4.54	3.85	0.64	0.51	0.69	2.45	3.7	2.85
HMM[acc]	2.26	5.24	4.02	1.68	3.16	10.91	8.68	9.89	9.0	2.12	3.66	3.35	10.55	12.6	10.19
HMM[acc+angles]	0.7	0.95	0.67	0.92	2.31	5.96	7.98	8.9	8.42	1.67	3.0	2.82	12.16	13.39	11.7

Since Table 2 showed that the HMMs seem to predict a lot more minutes spent in PA than the ENMO model and logbook, it is also interesting to see a more detailed comparison of this overestimation. Table 3 shows that on average both HMMs predict a similar amount of daily minutes in PA, but there are instances where the predictions are very far apart. The ENMO model predicts fewer daily minutes in PA compared to the HMMs, but is a lot closer to the reported minutes in PA.

Table 3: Average predicted minutes that were spent in physical activity over the 14 day period

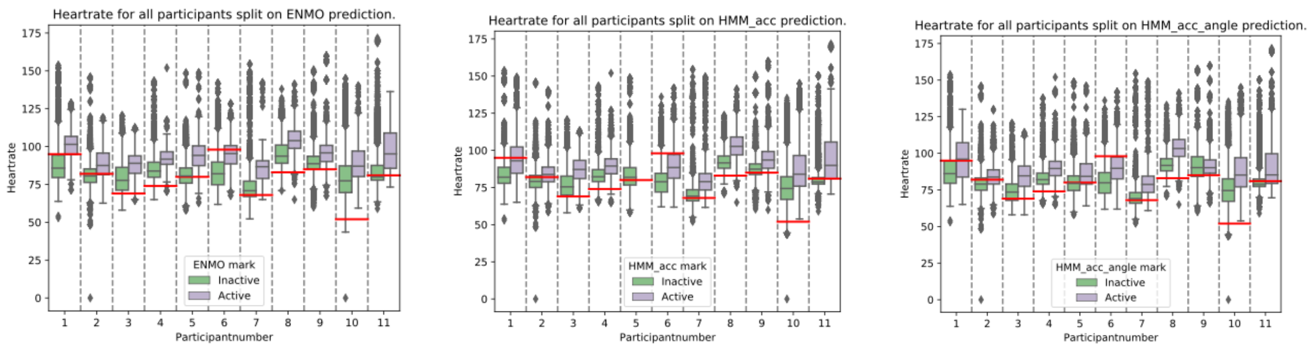
	1	2	3	4	5	6	7	8	9	10	11	Average
Logbook	146.94	0.0	0.0	0.0	0.0	135.46	0.0	29.32	0.0	145.98	22.76	43.68
ENMO model	71.8	44.42	105.78	45.0	110.66	89.14	18.37	168.63	111.43	85.8	71.41	83.85
HMM[acc]	569.63	263.51	269.09	272.38	0.0	424.76	237.03	342.95	256.79	483.82	292.61	310.32
HMM[acc+angles]	98.11	234.62	644.23	313.89	324.24	240.99	248.52	318.47	297.46	406.59	329.91	314.27

Table 4 reveals if these predicted activities are during the same timestamps as the reported activities. It shows the false negatives, the true positives, and the recall rate for each method and for all participants that reported any activities. The recall rate is a performance metric to check what method can more accurately predict a timestamp that was reported as active by a participant. Since those timestamps are very likely to actually have been spent in PA it is a good measure to see which model will be able to detect that activity. The recall rate is consistently higher for the HMMs than for the ENMO model. For every participant, the HMM[acc] also has a higher recall rate than HMM[acc+angles].

Table 4: Recall rates for the different models based on participant-reported activities

	1			6			8			10			11		
	FN	TP	Recall	FN	TP	Recall	FN	TP	Recall	FN	TP	Recall	FN	TP	Recall
ENMO model	4619	770	0.14	4639	661	0.12	1037	592	0.36	8168	1646	0.17	1222	464	0.28
HMM[acc]	1003	4386	0.81	1854	3446	0.65	243	1386	0.85	4087	5727	0.58	596	1090	0.65
HMM[acc+angles]	2360	3029	0.56	2826	2474	0.47	283	1346	0.83	4479	5335	0.54	627	1059	0.63

Figure 8 shows the spread of the heart rate for the different methods. Lower heart rates, especially below the measured heart rate at rest may indicate that someone was inactive during that time[23]. The red lines in Figure 8 correspond to the measured heart rate at rest at the hospital. It is visible that the medians of the inactive and active heart rates show more difference in the ENMO prediction than the HMM prediction. The lower boundary of the lower quartile also seems to be consistently higher for the ENMO model, meaning fewer measurements with lower heart rates were predicted as active.



(a) Split on ENMO prediction (b) Split on HMM[acc] prediction (c) Split on HMM[acc+angles] prediction

Figure 8: Comparison of the spread of heart rate data based on the prediction the model gave to that measurement

4 Discussion

This study evaluated the performance of two Hidden Markov Models utilizing different features in detecting physical activity from accelerometer data. Additionally, the performance of these models was compared to participant-reported activities and a more conventional cut-points approach. The two HMMs implemented in this research had two different observations as input. The first model (HMM[acc]) used the log-transformed ENMO values while the other model (HMM[acc+angles] used the log-transformed ENMO values as well as the angles of the acceleration. HMM[acc] showed a lot of dependence on the values the Gaussian emission distribution was initialised with, while HMM[acc+angles] was less reliant on the initialisation. A possible cause for this is that the several optima are all just equally as bad, because it fails to fit to the data of the angles. The goodness-of-fit is worse overall and the benefit of converging to an optima seemed to be less than for HMM[acc]. For HMM[acc] the initialisation was improved by performing a grid search around values found in literature[9]. This seemed to improve the consistency of the method, but in some instances it would still converge to a local maximum, which could result in a model with a suboptimal fit.

All methods showed the ability to predict activities of different durations and with a varying range of daily minutes in activity predicted (Figure 6). The ENMO model showed a lot more consistency in the duration of the predicted activities than the HMMs as can be seen in Figure 6, more than half of the activities predicted by the HMMs were shorter than 5 minutes. Activities shorter than 5 minutes were not possible for the ENMO model, which displays more consistency and reliability in estimating the duration of activities. Overall both HMMs predicted a lot more daily minutes spent in PA compared to the reported activities and the ENMO model. Table 2 also shows that the ENMO method fits the self-reported activities a lot better, because the RMSE overall is lower as well as the mean and median of the ratios of the daily predicted minutes and daily reported minutes. While this does imply that the HMMs predict a lot of inactive instances as active, comparing the heart rate data for the data split on the prediction showed that the spread of the heart rates do have a lot of similarities. The medians and lower quartiles were usually a fraction lower for the HMMs compared to the ENMO model. Implying there may be more inactive bouts during times activities were predicted for the HMMs than the ENMO model, or the HMMs are better at detecting lower-intensity activities. Since heart rates lower than the rest heart rate should not be possible during periods of physical activity. However, this measured rest heart rate might be higher than their rest heart rate in a relaxed day-to-day setting due to the white coat effect, which states that the rest heart rate measured in a clinical setting might be higher than the true heart rate at rest due to anxiety or stress during a clinic visit[24].

From the average recall rate it becomes evident that the HMMs were more accurate at detecting activities reported by the participants, the average recall rate was 0.22, 0.71 and 0.61 for the ENMO model, HMM[acc] and HMM[acc+angles] respectively. So during times where activities were reported by the participants, the HMMs were able to more adequately detect those reported activities than the ENMO model. However, Table 2 showed that the ENMO model overestimates the daily logbook reported minutes with only a factor 2 on average, while the HMMs did this with a factor of around 5 on average. The expected number of awake hours is around 16, this means that predicting 6+ hours of physical activity per day would mean 40 percent of the awake day was spent in physical activity. This is unrealistic, especially considering the population this research was conducted on. A possible explanation for the overestimation could be non-purposeful movements, such as fidgeting or arm movements while performing daily tasks, or even slight vibrations, that can be mistakenly classified as physical activity by the HMMs, because the accelerometer will measure a higher acceleration in such instances. Since the ENMO model is built to only predict activities longer than 5 minutes it is less sensitive to these temporary movements than the HMMs. The overestimation will give a large advantage for getting a higher recall rate, if you predict a lot more minutes spent in activities, chances are you will predict them during the reported bouts.

Comparing the two different feature sets for the Hidden Markov Model did not reveal large differences. The HMM[acc] demonstrated a higher recall rate while the other validation measures showed similar results as HMM[acc+angles]. However, this could be attributed to the fact that a better prior initialisation was found for HMM[acc]. The high recall rate of participant 1 is also caused by a large overestimation by HMM[acc]. HMM[acc] predicted 569.63 daily minutes of activity on average for participant 1 while HMM[acc+angles] predicted only 98.11 minutes. It is possible this also caused the difference in recall rate to be 25 percentage points (0.81 for HMM[acc] and 0.56 for HMM[acc+angles]). A possible explanation for this could be that the activities performed by participant 1 had a lot of similarity in its angle pattern thus allowing HMM[acc+angles] to recognize that pattern, but the ENMO values computed showed little difference between inactive periods and active periods. These extremely high predictions for daily minutes in activity were also visible for HMM[acc+angles] in participant 3, so additional research will be needed to (dis)prove that the angles improve the model's ability to detect active periods or if it will mostly be useful for classifying more specific activities[9][25].

4.1 Limitations

The ENMO model for this implementation did have its notable limitations. One of the limitations being the light physical activity threshold that was retrieved from a study in children and adults[13], which may not be optimal for a population with elderly participants. Another reason why the performance might be worse is due to the preprocessing step chosen, since the angles were smoothed using the same Gaussian filter as for the acceleration. Using the same Gaussian filter seemed logical to do so for consistency between the features, but after viewing the results it seemed that this had great consequences for the model. The angles might have been smoothed too much, which caused the consistent failure of meeting a threshold. For the general model it should hold for an activity that there is at least one occurrence where angle_z changed more than one degree per second between timestamps. If the values are smoothed too much the angle_z difference between timestamps becomes too small and the model will fail to find valid activities. A filter to remove outliers however still seems necessary, but these outliers are hard to attribute to just one factor. Measurement errors or rare events seem to be the most likely cause, but there still is a risk of losing valuable information when the outliers are smoothed away. Besides showing that the preprocessing filter was not optimal, this also shows the ENMO model's sensitivity to small changes and calibration errors.

The ENMO model as well as the HMMs showed to be sensitive to input which does not have the expected calibrated values. Most prominently noticeable in participant 7, where the ENMO model only predicted 20 active minutes per day, and the distribution of the data showed that a lot of ENMO values were calculated to be below zero 10. Further investigation revealed that the watch measured an acceleration in rest of around 9.6, which is significantly lower than the expected 9.81 (1g) which the ENMO calculation consistently accounts for. Causing a lot of ENMO values to be negative and set to zero. Which in this case made it too difficult for the HMMs to differentiate three states in the distribution and thus accurately predict the state, because the emission probabilities are assumed to be Gaussian distributed. This is currently a large flaw in the HMMs, the assumption of a Gaussian distribution of the features for different emission probabilities makes the model performance a lot worse if this distribution is not prominent. Assuming a Gaussian distribution in a simulation study did show the best results when classifying activity levels[14], but more research on real-life accelerometer data should be done. One way this assumed could be improved, is to use a different input than the ENMO values, for example just the magnitude of the acceleration. Or finding other physiologically relevant features that follow Gaussian distribution more than the current input.

Even though the model was initialised with the state-dependent expected means of the Gaussian distributed ENMO values, for participant 5 HMM[acc] still converged to an optimum where the means of state 1 and state 2 were very similar 5. This caused the model to have a very high chance of transitioning from state 2 to state 1, thus barely any active periods were predicted. Which also shows the risks of an HMM. Currently, after initialisation there is no control over what optimum the Baum-Welch algorithm converges to. It could be avoided by performing the Baum-Welch again until the ENMO values of the means of the Gaussian emission distribution are a certain distance apart.

For the ENMO model, the inconsistent sampling rate could be a big bottleneck for the performance. The requirement where in an activity 90 percent of the ENMO samples need to be above the threshold is chosen from literature and most literature works with a consistent sampling rate that is significantly higher. This threshold might be harder to reach with fewer measurements, since one inactive measurement could stop an activity. Setting a maximum of 1 minute with no measurements within an activity did not prevent this problem from occurring, because then the model failed to find any activities. Meaning that a lot of activities have gaps in the data. The HMMs showcase the same difficulties, since the model only works from observation to observation, it does not take into account how much time has passed within activities. Which could be the cause of extremely long activities visible in Figure 6. For smartwatches it is hard to prevent the inconsistent sampling rate, since setting a consistent sampling rate might negatively affect the lifetime of the battery and subsequently the wear-compliance.

A possible solution for this would be incorporating the pedometer data. As mentioned before, the watch will accumulate data during idle time. Currently, this is not taken into account but by analyzing the steps taken during a longer period without measurements it would be possible to detect if someone has been active in that period. Hidden Semi-Markov Models (HSMMs) could also help with preventing extremely long activities since they introduce the notion of duration modeling[9][2]. HSMMs have the ability to model the duration of a state and include an explicit distribution for the state duration. Whereas for HMMs the state duration is implicitly modeled through self-transitions. This means that HSMMs can represent the fact that certain states persist for longer or shorter periods of time, which makes the model more realistic.

In this research, for simplification, only three states were defined in the Hidden Markov models where one of the states was interpreted as active. This is done to keep it feasible within the timeframe this research was conducted. Changing the number of states to more closely model the health guidelines may improve the result significantly.

Another risk for the ENMO model as well as the HMMs, is that it could miss activities where the arms are relatively still. A wrist-worn accelerometer will not measure a large acceleration during those activities, while their legs might be very active. The detection of activity is improved considerably when working with multiple accelerometers or an accelerometer accompanied by another sensor, like heart rate, temperature, pressure, or GPS sensor[26]. For now this was not taken into account, as a higher heart rate does not necessarily suggest somebody is performing physical activity and this might unnecessarily complicate the model. More research looking into the types of activities that were missed could also show if there are specific types of activities that the model fails to detect and what measures would help to solve that.

4.2 Future work

Implementing the pedometer data as well as testing the HMMs for different numbers of states for the detection of physical activity is interesting for future research. Comparing this new model with a cut-points approach which is better at detecting physical activity might also bring new insights[27]. As well as testing the model for different features such as the impulse counts [14] and different preprocessing steps as this has mostly been done for other methods[28]. Post-processing could also improve the current model, some activities that are predicted are of very short duration, and this could easily be prevented. Potentially combining this with an implementation of HSMMs might also produce interesting as well as relevant results. Unsupervised learning approaches for activity detection from accelerometer data should be explored further in general, as it allows the model to discover patterns and structure in the data without relying on explicit labels or human-labeled training examples. It has the potential to contribute to the development of more accurate, adaptable, and interpretable activity detection systems.

5 Conclusion

In this study, Hidden Markov Models were implemented to detect physical activity from the smartwatch data of the participants. In contrast to other approaches like deep learning, the HMMs did not require large amounts of training data and complex architectures, thereby offering a simpler and more interpretable framework. Using an unsupervised learning approach completed negated the need for training data and lowered the chances of overfitting on population and sensor-specific data. When provided with physiologically meaningful features, HMMs demonstrated the ability to identify the correct hidden states. The implemented HMMs (HMM[acc] and HMM[acc+angles]) produced promising results with an average recall rate of 0.71 and 0.61 respectively, compared to 0.22 for the more standard cut-points approach. However, it was also shown that the method still requires additional work, since the results showed inconsistencies and some extremities. The HMMs predicted considerably higher numbers of active minutes compared to the ENMO model and the logbooks, which can bias the recall value and thus give a false impression of better performance in terms of activity detection. Still, the HMMs are a step in the direction of physical activity detection which can be performed without having the subjective bias of self-report methods or the need for calibration studies with a more standard cut-points approach. Allowing us to better understand what role physical activity plays regarding the health and disease of a person.

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A Appendix

A.1 Smoothed acceleration in y orientation

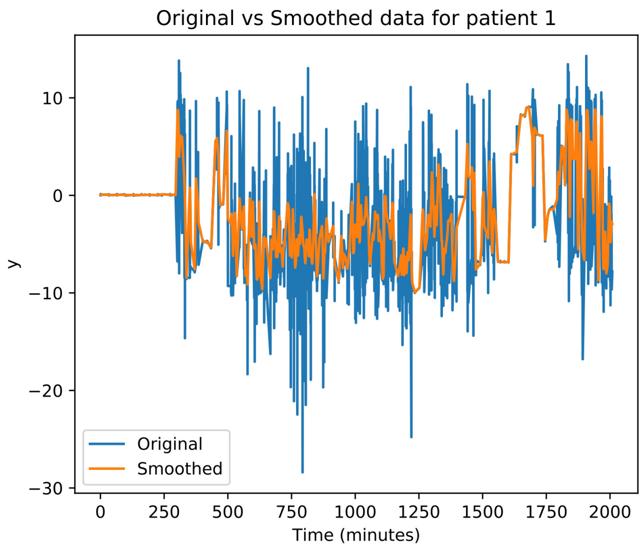
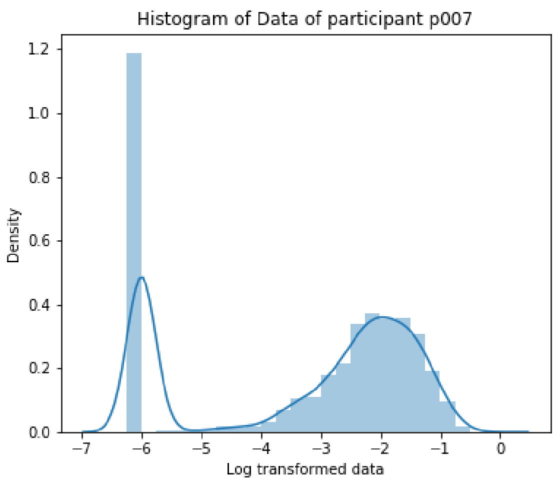


Figure 9: Original acceleration on y-axis plotted against smoothed acceleration

A.2 Log distributions of participants 1 and 7



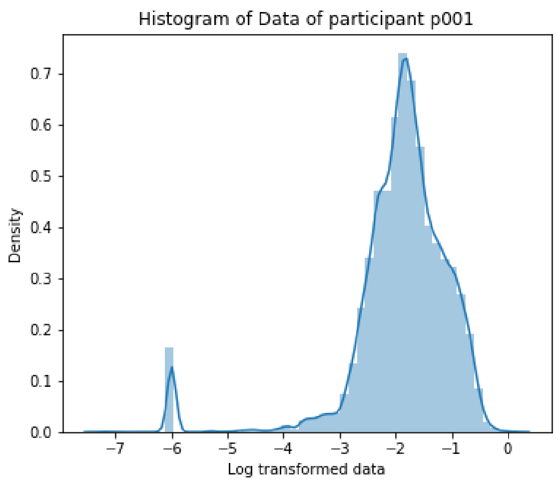


Figure 10: Logarithmic distribution of the ENMO values for participant 1 and 7

A.3 Fitscore for random initialization HMM[acc+angles]

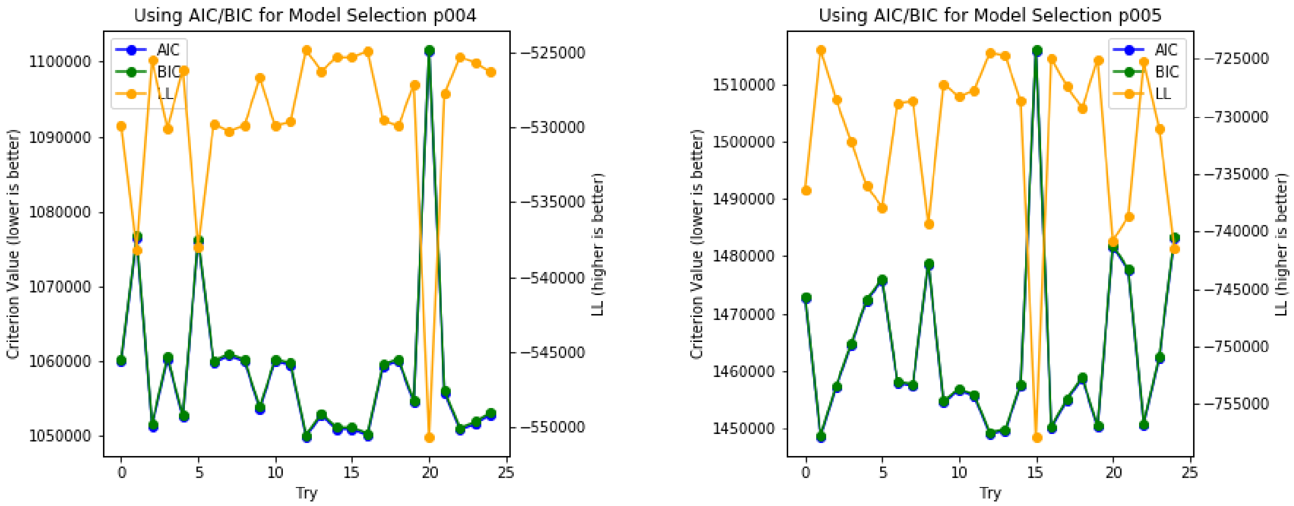


Figure 11: Goodness of fit for random initizaliation of HMM[acc+angles]

A.4 Parameters of Hidden Markov Models

Table 5: Parameters of the Hidden Markov Model using only the ENMO acceleration as feature.

Participants	Initial States	Transition Matrices	Means	Covars
1	$\begin{bmatrix}1.00\\0.00\\0.00\end{bmatrix}$	$\begin{bmatrix}0.96 & 0.02 & 0.02\\0.01 & 0.97 & 0.02\\0.00 & 0.01 & 0.98\end{bmatrix}$	$\begin{bmatrix}-3.51\\-2.13\\-1.33\end{bmatrix}$	$\begin{bmatrix}1.74\\0.04\\0.14\end{bmatrix}$
2	$\begin{bmatrix}1.00\\0.00\\0.00\end{bmatrix}$	$\begin{bmatrix}0.95 & 0.03 & 0.02\\0.02 & 0.96 & 0.03\\0.01 & 0.02 & 0.97\end{bmatrix}$	$\begin{bmatrix}-4.30\\-2.05\\-1.30\end{bmatrix}$	$\begin{bmatrix}2.22\\0.08\\0.05\end{bmatrix}$
3	$\begin{bmatrix}1.00\\0.00\\0.00\end{bmatrix}$	$\begin{bmatrix}0.95 & 0.04 & 0.01\\0.01 & 0.97 & 0.02\\0.01 & 0.04 & 0.96\end{bmatrix}$	$\begin{bmatrix}-3.92\\-1.69\\-1.03\end{bmatrix}$	$\begin{bmatrix}2.92\\0.05\\0.04\end{bmatrix}$
4	$\begin{bmatrix}1.00\\0.00\\0.00\end{bmatrix}$	$\begin{bmatrix}0.96 & 0.03 & 0.01\\0.01 & 0.97 & 0.02\\0.01 & 0.04 & 0.96\end{bmatrix}$	$\begin{bmatrix}-3.60\\-1.95\\-1.23\end{bmatrix}$	$\begin{bmatrix}1.61\\0.07\\0.07\end{bmatrix}$
5	$\begin{bmatrix}1.00\\0.00\\0.00\end{bmatrix}$	$\begin{bmatrix}0.97 & 0.00 & 0.03\\0.00 & 0.46 & 0.54\\0.03 & 0.94 & 0.02\end{bmatrix}$	$\begin{bmatrix}-2.77\\-1.43\\-1.44\end{bmatrix}$	$\begin{bmatrix}0.74\\0.12\\0.13\end{bmatrix}$
6	$\begin{bmatrix}0.00\\1.00\\0.00\end{bmatrix}$	$\begin{bmatrix}0.94 & 0.03 & 0.03\\0.01 & 0.97 & 0.02\\0.00 & 0.02 & 0.97\end{bmatrix}$	$\begin{bmatrix}-3.45\\-2.15\\-1.30\end{bmatrix}$	$\begin{bmatrix}1.43\\0.06\\0.08\end{bmatrix}$
7	$\begin{bmatrix}1.00\\0.00\\0.00\end{bmatrix}$	$\begin{bmatrix}0.96 & 0.03 & 0.01\\0.02 & 0.96 & 0.01\\0.01 & 0.03 & 0.96\end{bmatrix}$	$\begin{bmatrix}-5.77\\-2.42\\-1.38\end{bmatrix}$	$\begin{bmatrix}0.41\\0.22\\0.08\end{bmatrix}$
8	$\begin{bmatrix}1.00\\0.00\\0.00\end{bmatrix}$	$\begin{bmatrix}0.97 & 0.02 & 0.01\\0.01 & 0.97 & 0.02\\0.00 & 0.02 & 0.98\end{bmatrix}$	$\begin{bmatrix}-3.97\\-1.80\\-0.95\end{bmatrix}$	$\begin{bmatrix}1.55\\0.07\\0.06\end{bmatrix}$
9	$\begin{bmatrix}0.00\\1.00\\0.00\end{bmatrix}$	$\begin{bmatrix}0.97 & 0.02 & 0.02\\0.00 & 0.98 & 0.02\\0.00 & 0.02 & 0.98\end{bmatrix}$	$\begin{bmatrix}-2.47\\-1.77\\-1.13\end{bmatrix}$	$\begin{bmatrix}0.93\\0.03\\0.04\end{bmatrix}$
10	$\begin{bmatrix}0.00\\1.00\\0.00\end{bmatrix}$	$\begin{bmatrix}0.99 & 0.01 & 0.00\\0.00 & 0.98 & 0.02\\0.00 & 0.02 & 0.98\end{bmatrix}$	$\begin{bmatrix}-5.53\\-2.15\\-1.26\end{bmatrix}$	$\begin{bmatrix}0.88\\0.13\\0.08\end{bmatrix}$
11	$\begin{bmatrix}0.00\\1.00\\0.00\end{bmatrix}$	$\begin{bmatrix}0.99 & 0.01 & 0.00\\0.00 & 0.98 & 0.02\\0.00 & 0.02 & 0.97\end{bmatrix}$	$\begin{bmatrix}-5.20\\-1.93\\-1.15\end{bmatrix}$	$\begin{bmatrix}1.49\\0.06\\0.08\end{bmatrix}$

Table 6: Parameters of the HMM+acc+angles

Participants	Initial States	Transition Matrices	Means	Covars
1	$\begin{bmatrix}1.00 & 0.00 & 0.00\end{bmatrix}$	$\begin{bmatrix}0.98 & 0.00 & 0.02\\ 0.00 & 0.00 & 1.00\\ 0.01 & 0.43 & 0.56\end{bmatrix}$	$\begin{bmatrix}-2.49 & 4.49 & 24.36 & 37.46\\ -1.50 & -3.19 & -30.97 & 31.84\\ -1.47 & -2.48 & -31.23 & 29.01\end{bmatrix}$	$\begin{bmatrix}\begin{bmatrix}1.12 & 0.00 & 0.00 & 0.00\\ 0.00 & 339.16 & 0.00 & 0.00\\ 0.00 & 0.00 & 805.75 & 0.00\\ 0.00 & 0.00 & 0.00 & 1362.83\end{bmatrix}\\ \begin{bmatrix}0.21 & 0.00 & 0.00 & 0.00\\ 0.00 & 440.15 & 0.00 & 0.00\\ 0.00 & 0.00 & 278.95 & 0.00\\ 0.00 & 0.00 & 0.00 & 857.20\end{bmatrix}\\ \begin{bmatrix}0.22 & 0.00 & 0.00 & 0.00\\ 0.00 & 493.39 & 0.00 & 0.00\\ 0.00 & 0.00 & 279.91 & 0.00\\ 0.00 & 0.00 & 0.00 & 929.70\end{bmatrix}\end{bmatrix}$
2	$\begin{bmatrix}1.00 & 0.00 & 0.00\end{bmatrix}$	$\begin{bmatrix}0.96 & 0.03 & 0.02\\ 0.02 & 0.96 & 0.03\\ 0.01 & 0.02 & 0.97\end{bmatrix}$	$\begin{bmatrix}-3.85 & 22.45 & 3.13 & 44.60\\ -2.00 & 7.93 & -14.45 & 45.15\\ -1.33 & 2.53 & -15.56 & 47.94\end{bmatrix}$	$\begin{bmatrix}\begin{bmatrix}2.67 & 0.00 & 0.00 & 0.00\\ 0.00 & 422.46 & 0.00 & 0.00\\ 0.00 & 0.00 & 828.78 & 0.00\\ 0.00 & 0.00 & 0.00 & 535.46\end{bmatrix}\\ \begin{bmatrix}0.08 & 0.00 & 0.00 & 0.00\\ 0.00 & 318.29 & 0.00 & 0.00\\ 0.00 & 0.00 & 820.60 & 0.00\\ 0.00 & 0.00 & 0.00 & 467.34\end{bmatrix}\\ \begin{bmatrix}0.07 & 0.00 & 0.00 & 0.00\\ 0.00 & 154.30 & 0.00 & 0.00\\ 0.00 & 0.00 & 211.86 & 0.00\\ 0.00 & 0.00 & 0.00 & 129.55\end{bmatrix}\end{bmatrix}$
3	$\begin{bmatrix}1.00 & 0.00 & 0.00\end{bmatrix}$	$\begin{bmatrix}0.95 & 0.04 & 0.02\\ 0.02 & 0.95 & 0.03\\ 0.00 & 0.01 & 0.98\end{bmatrix}$	$\begin{bmatrix}-4.21 & 34.27 & 23.57 & 10.40\\ -1.88 & 2.68 & -2.48 & 37.48\\ -1.30 & -11.53 & -28.38 & 38.50\end{bmatrix}$	$\begin{bmatrix}\begin{bmatrix}3.25 & 0.00 & 0.00 & 0.00\\ 0.00 & 979.93 & 0.00 & 0.00\\ 0.00 & 0.00 & 580.82 & 0.00\\ 0.00 & 0.00 & 0.00 & 1010.03\end{bmatrix}\\ \begin{bmatrix}0.11 & 0.00 & 0.00 & 0.00\\ 0.00 & 285.59 & 0.00 & 0.00\\ 0.00 & 0.00 & 1213.41 & 0.00\\ 0.00 & 0.00 & 0.00 & 780.73\end{bmatrix}\\ \begin{bmatrix}0.12 & 0.00 & 0.00 & 0.00\\ 0.00 & 346.60 & 0.00 & 0.00\\ 0.00 & 0.00 & 251.59 & 0.00\\ 0.00 & 0.00 & 0.00 & 269.57\end{bmatrix}\end{bmatrix}$
4	$\begin{bmatrix}1.00 & 0.00 & 0.00\end{bmatrix}$	$\begin{bmatrix}0.963 & 0.025 & 0.011\\ 0.012 & 0.966 & 0.022\\ 0.007 & 0.030 & 0.964\end{bmatrix}$	$\begin{bmatrix}-3.53 & 5.24 & 6.94 & 53.74\\ -1.97 & 2.29 & -12.35 & 39.41\\ -1.29 & -8.51 & -20.03 & 35.05\end{bmatrix}$	$\begin{bmatrix}\begin{bmatrix}1.65 & 0.00 & 0.00 & 0.00\\ 0.00 & 258.41 & 0.00 & 0.00\\ 0.00 & 0.00 & 400.95 & 0.00\\ 0.00 & 0.00 & 0.00 & 1849.32\end{bmatrix}\\ \begin{bmatrix}0.07 & 0.00 & 0.00 & 0.00\\ 0.00 & 251.40 & 0.00 & 0.00\\ 0.00 & 0.00 & 812.09 & 0.00\\ 0.00 & 0.00 & 0.00 & 1212.38\end{bmatrix}\\ \begin{bmatrix}0.09 & 0.00 & 0.00 & 0.00\\ 0.00 & 501.17 & 0.00 & 0.00\\ 0.00 & 0.00 & 286.66 & 0.00\\ 0.00 & 0.00 & 0.00 & 400.44\end{bmatrix}\end{bmatrix}$

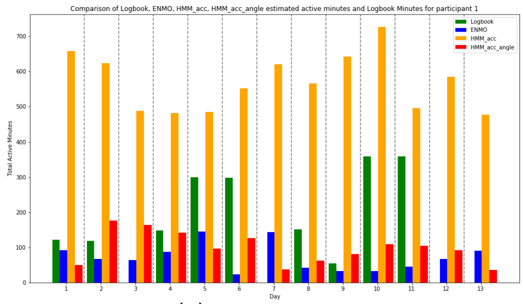
Table 7: Parameters of the Hidden Markov Model using the ENMO acceleration as well as angles as features.

Participants	Initial States	Transition Matrices	Means	Covars
5	$\begin{bmatrix}1.00 & 0.00 & 0.00\end{bmatrix}$	$\begin{bmatrix}0.969 & 0.015 & 0.016\\0.009 & 0.964 & 0.026\\0.005 & 0.017 & 0.978\end{bmatrix}$	$\begin{bmatrix}-3.10 & -2.24 & -14.12 & 61.62\\-1.92 & 4.03 & -16.81 & 9.67\\-1.35 & -5.96 & -16.99 & 42.06\end{bmatrix}$	$\begin{bmatrix}\begin{bmatrix}0.96 & 0.00 & 0.00 & 0.00\\0.00 & 89.45 & 0.00 & 0.00\\0.00 & 0.00 & 433.71 & 0.00\\0.00 & 0.00 & 0.00 & 2341.13\end{bmatrix}\\\begin{bmatrix}0.05 & 0.00 & 0.00 & 0.00\\0.00 & 422.77 & 0.00 & 0.00\\0.00 & 0.00 & 737.94 & 0.00\\0.00 & 0.00 & 0.00 & 952.35\end{bmatrix}\\\begin{bmatrix}0.07 & 0.00 & 0.00 & 0.00\\0.00 & 490.84 & 0.00 & 0.00\\0.00 & 0.00 & 289.95 & 0.00\\0.00 & 0.00 & 0.00 & 400.62\end{bmatrix}\end{bmatrix}$
6	$\begin{bmatrix}1.00 & 0.00 & 0.00\end{bmatrix}$	$\begin{bmatrix}0.967 & 0.021 & 0.012\\0.011 & 0.970 & 0.019\\0.006 & 0.028 & 0.966\end{bmatrix}$	$\begin{bmatrix}-3.44 & -1.13 & -17.47 & 28.91\\-2.27 & -3.56 & -12.11 & 39.62\\-1.23 & -4.36 & -20.45 & 29.34\end{bmatrix}$	$\begin{bmatrix}\begin{bmatrix}1.47 & 0.00 & 0.00 & 0.00\\0.00 & 278.11 & 0.00 & 0.00\\0.00 & 0.00 & 424.23 & 0.00\\0.00 & 0.00 & 0.00 & 768.81\end{bmatrix}\\\begin{bmatrix}0.04 & 0.00 & 0.00 & 0.00\\0.00 & 327.68 & 0.00 & 0.00\\0.00 & 0.00 & 546.33 & 0.00\\0.00 & 0.00 & 0.00 & 1065.84\end{bmatrix}\\\begin{bmatrix}0.07 & 0.00 & 0.00 & 0.00\\0.00 & 528.62 & 0.00 & 0.00\\0.00 & 0.00 & 351.74 & 0.00\\0.00 & 0.00 & 0.00 & 816.94\end{bmatrix}\end{bmatrix}$
7	$\begin{bmatrix}1.00 & 6.78 \times 10^{-211} & 0.00\end{bmatrix}$	$\begin{bmatrix}0.96 & 0.03 & 0.01\\0.02 & 0.96 & 0.01\\0.01 & 0.02 & 0.96\end{bmatrix}$	$\begin{bmatrix}-6.00 & 5.08 & -10.83 & 70.23\\-2.58 & 7.84 & -19.19 & 44.28\\-1.47 & -9.23 & -24.79 & 41.23\end{bmatrix}$	$\begin{bmatrix}\begin{bmatrix}1.28 \times 10^{-6} & 0.00 & 0.00 & 0.00\\0.00 & 218.51 & 0.00 & 0.00\\0.00 & 0.00 & 425.38 & 0.00\\0.00 & 0.00 & 0.00 & 688.96\end{bmatrix}\\\begin{bmatrix}0.48 & 0.00 & 0.00 & 0.00\\0.00 & 194.32 & 0.00 & 0.00\\0.00 & 0.00 & 820.13 & 0.00\\0.00 & 0.00 & 0.00 & 834.99\end{bmatrix}\\\begin{bmatrix}0.12 & 0.00 & 0.00 & 0.00\\0.00 & 334.42 & 0.00 & 0.00\\0.00 & 0.00 & 246.85 & 0.00\\0.00 & 0.00 & 0.00 & 301.44\end{bmatrix}\end{bmatrix}$
8	$\begin{bmatrix}1.00 & 9.66 \times 10^{-63} & 1.91 \times 10^{-173}\end{bmatrix}$	$\begin{bmatrix}0.97 & 0.02 & 0.01\\0.01 & 0.97 & 0.02\\0.00 & 0.02 & 0.98\end{bmatrix}$	$\begin{bmatrix}-3.90 & 3.16 & 1.86 & 71.90\\-1.74 & -3.29 & -12.99 & 48.48\\-0.93 & -11.12 & -26.30 & 32.21\end{bmatrix}$	$\begin{bmatrix}\begin{bmatrix}1.61 & 0.00 & 0.00 & 0.00\\0.00 & 432.88 & 0.00 & 0.00\\0.00 & 0.00 & 154.97 & 0.00\\0.00 & 0.00 & 0.00 & 751.50\end{bmatrix}\\\begin{bmatrix}0.09 & 0.00 & 0.00 & 0.00\\0.00 & 314.63 & 0.00 & 0.00\\0.00 & 0.00 & 475.17 & 0.00\\0.00 & 0.00 & 0.00 & 532.97\end{bmatrix}\\\begin{bmatrix}0.06 & 0.00 & 0.00 & 0.00\\0.00 & 303.52 & 0.00 & 0.00\\0.00 & 0.00 & 165.06 & 0.00\\0.00 & 0.00 & 0.00 & 245.05\end{bmatrix}\end{bmatrix}$

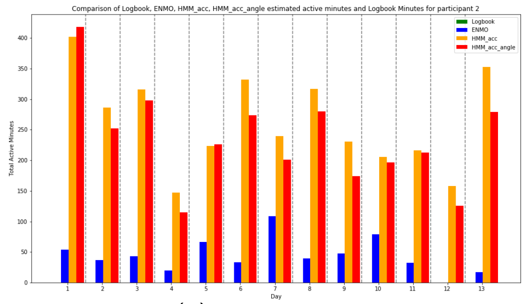
Table 8: Parameters of the Hidden Markov Model using the ENMO acceleration as well as angles as features.

Participants	Initial states	Transition Matrix	Means	Covariance Matrix
9	$\begin{bmatrix}0.00 & 1.00 & 0.00\end{bmatrix}$	$\begin{bmatrix}0.97 & 0.00 & 0.03\\0.00 & 0.99 & 0.00\\0.02 & 0.00 & 0.98\end{bmatrix}$	$\begin{bmatrix}-1.52 & -4.60 & -21.87 & 30.76\\-1.89 & -0.57 & -0.60 & -24.75\\-1.40 & -4.12 & -16.62 & 58.25\end{bmatrix}$	$\begin{bmatrix}\begin{bmatrix}0.65 & 0.00 & 0.00 & 0.00\\0.00 & 437.49 & 0.00 & 0.00\\0.00 & 0.00 & 853.32 & 0.00\\0.00 & 0.00 & 0.00 & 633.70\end{bmatrix}\\\begin{bmatrix}0.05 & 0.00 & 0.00 & 0.00\\0.00 & 0.56 & 0.00 & 0.00\\0.00 & 0.00 & 0.83 & 0.00\\0.00 & 0.00 & 0.00 & 7101.14\end{bmatrix}\\\begin{bmatrix}0.12 & 0.00 & 0.00 & 0.00\\0.00 & 68.99 & 0.00 & 0.00\\0.00 & 0.00 & 93.40 & 0.00\\0.00 & 0.00 & 0.00 & 103.75\end{bmatrix}\end{bmatrix}$
10	$\begin{bmatrix}0.00 & 1.00 & 0.00\end{bmatrix}$	$\begin{bmatrix}0.99 & 0.01 & 0.00\\0.00 & 0.98 & 0.01\\0.00 & 0.02 & 0.98\end{bmatrix}$	$\begin{bmatrix}-5.62 & 4.53 & -0.43 & 81.21\\-2.10 & 1.21 & -23.15 & 43.70\\-1.24 & -9.61 & -27.29 & 40.45\end{bmatrix}$	$\begin{bmatrix}\begin{bmatrix}0.71 & 0.00 & 0.00 & 0.00\\0.00 & 333.22 & 0.00 & 0.00\\0.00 & 0.00 & 51.52 & 0.00\\0.00 & 0.00 & 0.00 & 438.38\end{bmatrix}\\\begin{bmatrix}0.20 & 0.00 & 0.00 & 0.00\\0.00 & 245.02 & 0.00 & 0.00\\0.00 & 0.00 & 758.31 & 0.00\\0.00 & 0.00 & 0.00 & 930.38\end{bmatrix}\\\begin{bmatrix}0.09 & 0.00 & 0.00 & 0.00\\0.00 & 277.68 & 0.00 & 0.00\\0.00 & 0.00 & 130.56 & 0.00\\0.00 & 0.00 & 0.00 & 256.58\end{bmatrix}\end{bmatrix}$
11	$\begin{bmatrix}0.00 & 1.00 & 0.00\end{bmatrix}$	$\begin{bmatrix}0.99 & 0.00 & 0.00\\0.00 & 0.98 & 0.02\\0.00 & 0.02 & 0.98\end{bmatrix}$	$\begin{bmatrix}-5.46 & -0.56 & 1.65 & 85.88\\-2.07 & -9.35 & -24.07 & 29.95\\-1.26 & -11.23 & -20.48 & 43.72\end{bmatrix}$	$\begin{bmatrix}\begin{bmatrix}1.19 & 0.00 & 0.00 & 0.00\\0.00 & 9.31 & 0.00 & 0.00\\0.00 & 0.00 & 18.26 & 0.00\\0.00 & 0.00 & 0.00 & 21.03\end{bmatrix}\\\begin{bmatrix}0.41 & 0.00 & 0.00 & 0.00\\0.00 & 366.83 & 0.00 & 0.00\\0.00 & 0.00 & 1123.16 & 0.00\\0.00 & 0.00 & 0.00 & 983.54\end{bmatrix}\\\begin{bmatrix}0.13 & 0.00 & 0.00 & 0.00\\0.00 & 274.39 & 0.00 & 0.00\\0.00 & 0.00 & 142.94 & 0.00\\0.00 & 0.00 & 0.00 & 231.36\end{bmatrix}\end{bmatrix}$

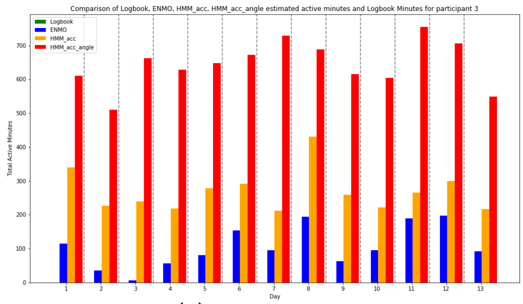
A.5 Plots for predicted daily minutes spent in PA



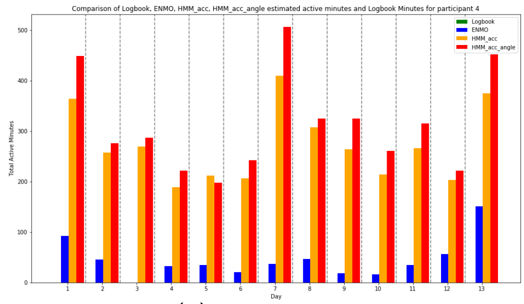
(a) Participant 1



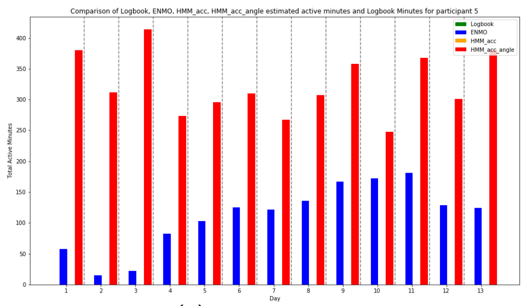
(b) Participant 2



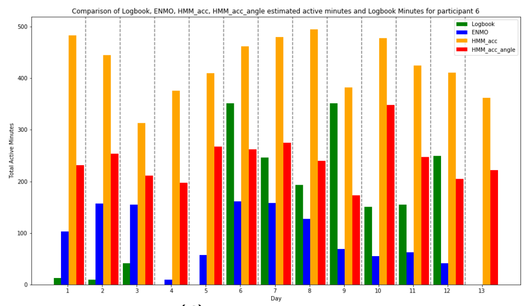
(c) Participant 3



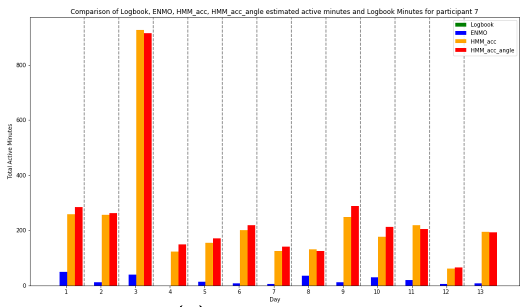
(d) Participant 4



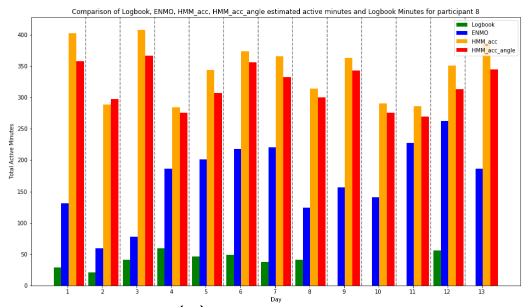
(e) Participant 5



(f) Participant 6

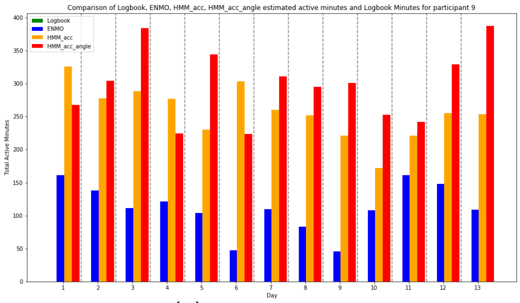


(g) Participant 7

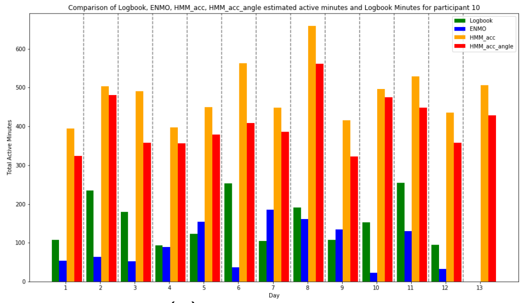


(h) Participant 8

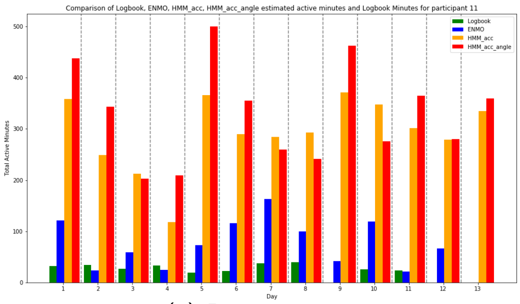
Figure 12: Plots for predicted daily minutes spent in PA



(a) Participant 9



(b) Participant 10



(c) Participant 11

Figure 13: Plots for predicted daily minutes spent in PA